

The Cross Sectional Multilayer Perceptron for Portfolio Construction

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Abstract

This document describes the cross sectional multilayer perceptron for monthly portfolio construction. The model encodes each security from its characteristics, mixes that encoding with a pooled summary of the whole cross section so that securities inform one another before being scored, and is fitted under a maximum Sharpe ratio regression objective. We set out the theoretical framework in full, namely the representation of the monthly cross section, the objective, and the pooled cross sectional coupling that distinguishes the model from a per stock forecaster. We then describe the empirical protocol, that is the rolling window, the per seed warm starting, and the seed averaging, before reporting the realised performance on the evaluation window.

1 Overview and motivation

In every month the task is to supply a vector of portfolio weights over the securities present in that month, and the resulting portfolio is scored by the annualised Sharpe ratio of its monthly return series over the test window from December 1989 to December 2023. The task is therefore not one of forecasting a return for each security in isolation, but one of assigning a relative weight to each security conditional on the composition of the cross section in which it sits.

This distinction motivates the central design choice. A model that maps a single security's characteristics to a score, applied identically to every security, cannot represent the fact that the attractiveness of a given profile depends on the other securities available in the same month. We therefore adopt an architecture in which each security is first encoded independently and is then mixed with a pooled summary of the whole cross section before it is scored. This pooled coupling is the mechanism that supplies the cross sectional signal that a per stock model misses, and the evidence on this panel indicates that it is the single most important architectural feature of the model.

2 Data representation and preprocessing

The raw inputs are a panel of monthly security characteristics and a list of candidate feature names, and the choice of representation determines what the network is able to learn. The prediction target is the excess return of a security over the month following the formation date, denoted r and stored in the field `ret_exc_lead1m`.

The candidate features are first intersected with the columns actually present in the characteristics panel. Only those features whose coverage, that is the fraction of non missing observations, is at least 0.90 over the pre test period ending in January 1990 are retained. This coverage filter is computed only from pre test data and is therefore causal, and it is performed programmatically,

which satisfies the requirement that feature selection be algorithmic rather than based on known historical performance.

The input matrix for each formation month is formed as follows. Every retained characteristic is transformed to its within month percentile rank, so that the value supplied to the network is the cross sectional rank of the security on that characteristic rather than its level. Let us define the transformed value of security i on characteristic j as

$$x_{ij} = \text{rank}_j(i) - 0.5, \quad (1)$$

where $\text{rank}_j(i) \in (0, 1]$ is the percentile rank within the month and missing entries are assigned the median value 0.5 before centring. The resulting inputs lie in the interval $[-0.5, 0.5]$ and are free of scale, which makes the representation robust to the heavy tails and changing dispersion of raw accounting and market variables. Securities missing more than one third of the retained characteristics are dropped, and a month is retained only if at least thirty securities survive.

3 Architecture

The network consists of a per stock encoder, a pooled context, and a scoring head. The encoder is a two layer perceptron with layer normalisation and a Gaussian error linear unit nonlinearity that maps the characteristic vector of each security to an embedding $z_i \in \mathbb{R}^d$ with $d = 128$,

$$z_i = \text{encoder}(x_i). \quad (2)$$

The pooled context is the average embedding across all securities present in the month,

$$c = \frac{1}{n} \sum_{i=1}^n z_i, \quad (3)$$

and it is the same vector for every security in that month. The score is produced by a head that receives the concatenation of the security embedding and the pooled context,

$$s_i = \text{head}([z_i, c]). \quad (4)$$

Because c depends on the whole cross section, the score of a security is a function not only of its own profile but of how that profile compares with the rest of the month. We can observe that in the absence of the context term the model would reduce to a per stock forecaster applied independently, and the evidence that removing this coupling reduces the Sharpe ratio substantially indicates that the term is doing genuine work.

4 Training objective

In a given training month let $w = s$ denote the vector of raw scores and r the vector of realised excess returns. The loss for that month is

$$\ell = \left(1 - \sum_i w_i r_i\right)^2, \quad (5)$$

and the total loss is the average of ℓ over the training months. This is the maximum Sharpe ratio regression objective. It is the sample analogue of regressing a vector of ones on the security returns

without an intercept, whose solution is proportional to $\Sigma^{-1}\mu$, the tangency portfolio of mean variance theory. By driving the realised managed return $\sum_i w_i r_i$ towards unity in every month, the objective simultaneously rewards a high average portfolio return and penalises its variability, which is precisely the trade off summarised by the Sharpe ratio. The network learns a single nonlinear map from characteristics to tangency style weights that is shared across all months.

The scores are trained without any normalisation, so their scale is free during optimisation. The traded weights are obtained at prediction time by normalising the scores to unit gross exposure,

$$\tilde{w}_i = \frac{w_i}{\sum_j |w_j|}, \quad (6)$$

which fixes the leverage of the book and leaves the cross sectional pattern of the weights unchanged. Since the Sharpe ratio is invariant to this scaling, the normalisation affects reported gross exposure but not the score.

5 Rolling estimation and temporal integrity

The estimation protocol is designed so that no information from the future can enter the weights at any formation date. For each formation month we take the trailing window of at most sixty months whose forward returns are already observable at that date, and we fit the network on those months only. The first time a seed is fitted it is trained from a random initialisation for fifty epochs, which we refer to as a cold start. At each subsequent month the seed is warm started from its own previous state and trained for a further ten epochs, so that the network adapts to the most recent window while retaining what it learned earlier. This warm starting reduces the cost of monthly refitting and stabilises the estimates.

To reduce the dependence of the result on the random initialisation, three independent seeds are fitted. For each seed the out of sample scores at the formation month are normalised to unit gross exposure, and the normalised weights are averaged across the seeds to give the traded book. All of these operations use only data available at or before the formation date, so the construction respects temporal integrity.

6 Evaluation

The book is evaluated by forming the monthly portfolio return $R_t = \sum_i \tilde{w}_{it} r_{it}$ over the months whose forward returns are observed, and computing the annualised Sharpe ratio

$$\text{SR} = \frac{\bar{R}}{\hat{\sigma}_R} \sqrt{12}, \quad (7)$$

where $\bar{R}$ and $\hat{\sigma}_R$ are the sample mean and standard deviation of the monthly returns. A volatility scaled variant is also reported, in which the book is levered each month by the ratio of a fixed annual volatility target of ten percent to a trailing thirty six month estimate of realised volatility, with the leverage lagged so that it uses only past information and clipped to the interval $[0, 10]$. This overlay changes the time path of leverage but not the underlying weights, and because the Sharpe ratio is scale invariant, any improvement it produces reflects the timing of leverage rather than any change in security selection.

On the evaluation window the model attains a Sharpe ratio of 2.484 in the volatility scaled form and 2.553 in the unscaled form.

7 Concluding remarks

The cross sectional multilayer perceptron demonstrates that a compact network, trained under a maximum Sharpe ratio regression objective on ranked characteristics and coupled across the cross section through a single pooled context, produces a competitive portfolio on this panel. The result suggests that the pooled coupling, rather than the specific form of the encoder, is the feature that carries the performance, and this observation motivates the natural extensions that replace the pooled context with richer forms of cross sectional interaction, that condition the map on the market state, or that add an explicit risk model on top of the selection signal. Within the limitations of the present analysis, the model is a reliable and interpretable construction that stands on its own and serves as a firm point of departure.